IoT Anomaly Detection

Industrial Sensor Monitoring with Statistical & ML Methods

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Agenda

1. Project Overview
2. Sensor Data
3. Statistical Methods
4. ML Methods
5. Detection & Tuning
6. Root Cause
7. Conclusions

Industrial IoT Monitoring

Why It Matters

Industrial machines (pumps, motors, turbines) are instrumented with sensors to enable:

- **Predictive maintenance** — catch failures early
- **Safety** — alert on dangerous operating conditions
- **Efficiency** — detect deviations from optimal performance
- **Quality control** — flag process drift

Setup

Duration	30 days
Sampling	1 minute
Total samples	43,200
Sensors	5
Anomaly events	47
Anomaly minutes	2,000+

Approach

Compare 7 anomaly detection methods on 30 days

Sensors & Anomaly Types

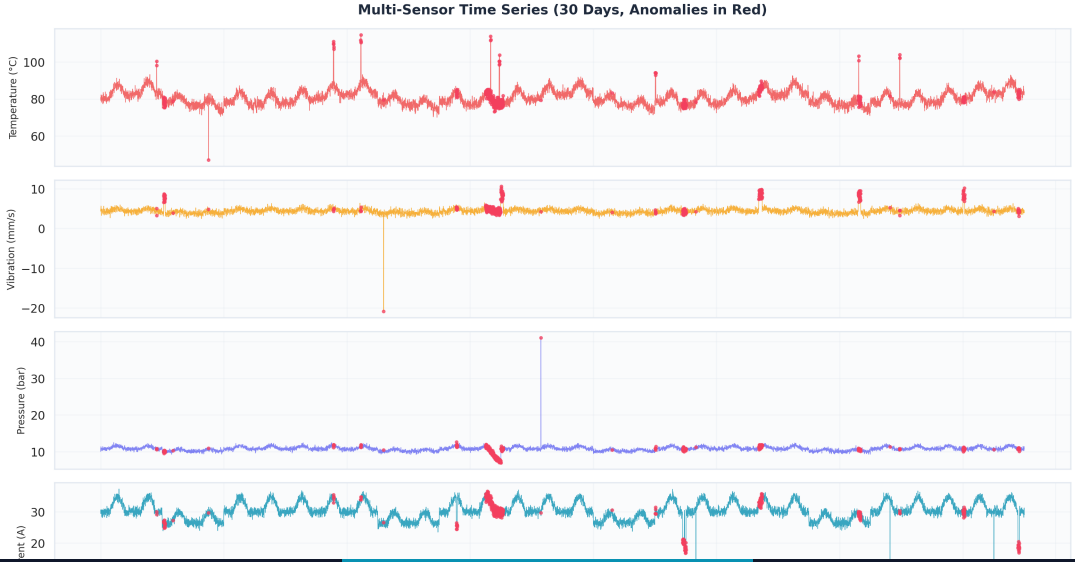
5 Sensors Monitored

- Temperature (°C)
- Vibration (mm/s RMS)
- Pressure (bar)
- Current draw (Amps)
- Flow rate (L/min)

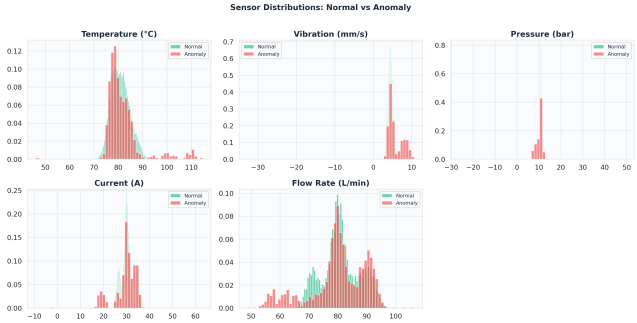
5 Anomaly Types Injected

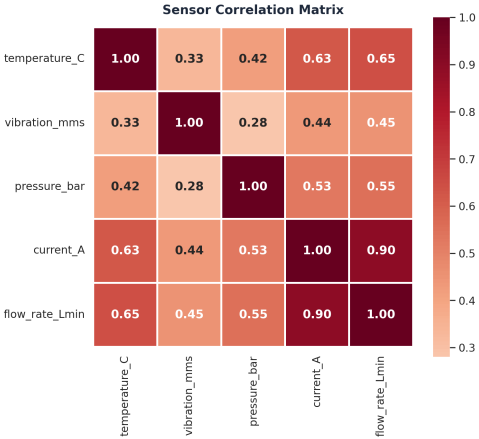
1. Temperature spikes (overheating)
2. High vibration (bearing wear)
3. Pressure drift (developing leak)
4. Multi-sensor failure (current+flow)
5. Sensor glitches (brief outliers)

Multi-Sensor Time Series (30 Days)



Sensor Distributions & Correlations

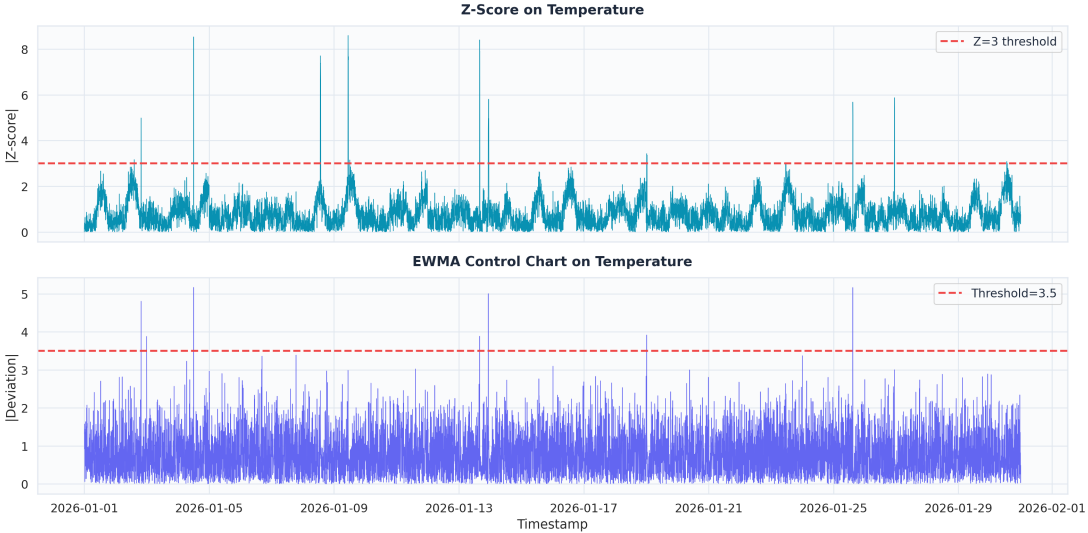




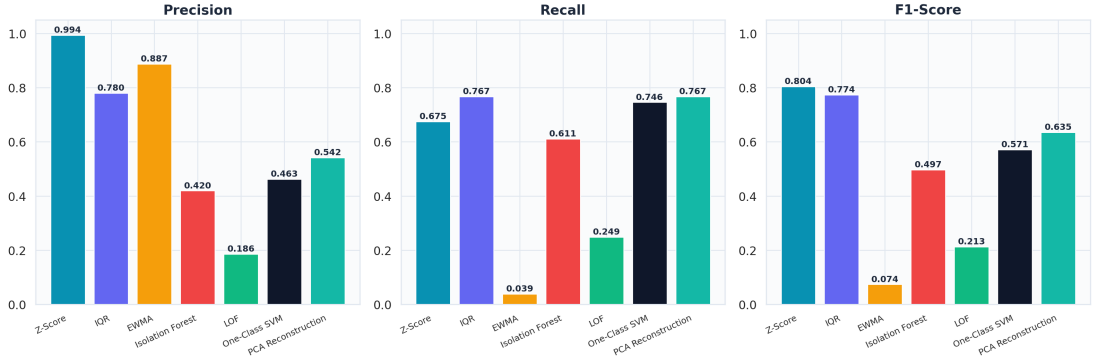
Pattern Recognition

Anomaly distributions extend into the tails of normal operation. The correlation matrix shows

Z-Score & EWMA Control Charts



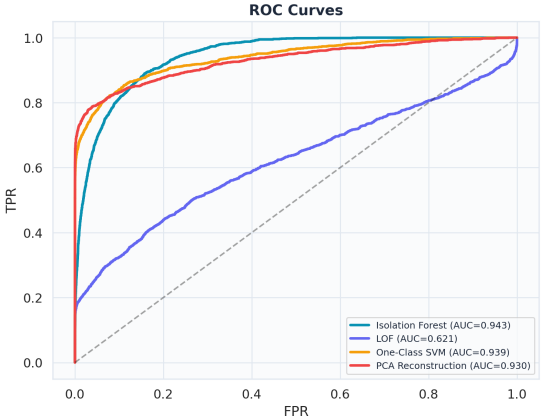
Method Comparison (Precision / Recall / F1)

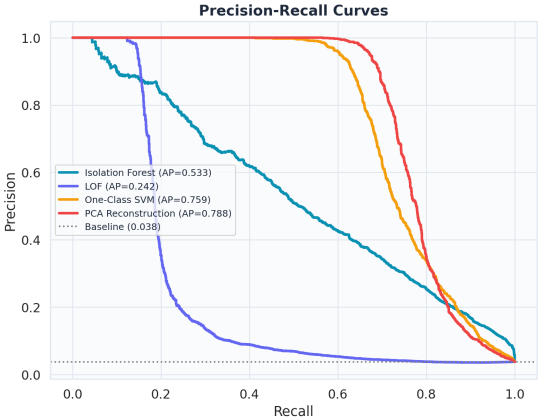


Highlights

- **Z-Score** achieves **99% precision** with strict threshold (very few false alarms)
- **PCA Reconstruction** balances P/R best (F1 = 0.64)
- **LOF** struggles — multivariate temporal data isn't its sweet spot

ROC & Precision-Recall Curves

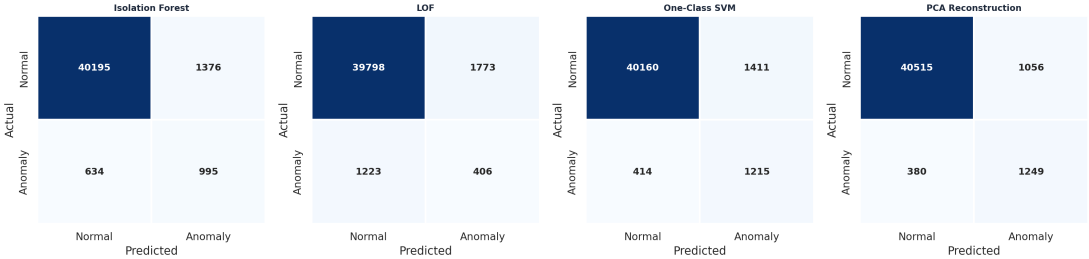




Ranking Quality

PCA Reconstruction and One-Class SVM both achieve **AUC ¿ 0.93** — excellent for ranking-based alert prioritization. AUC-PR is the more honest metric given the rare-event

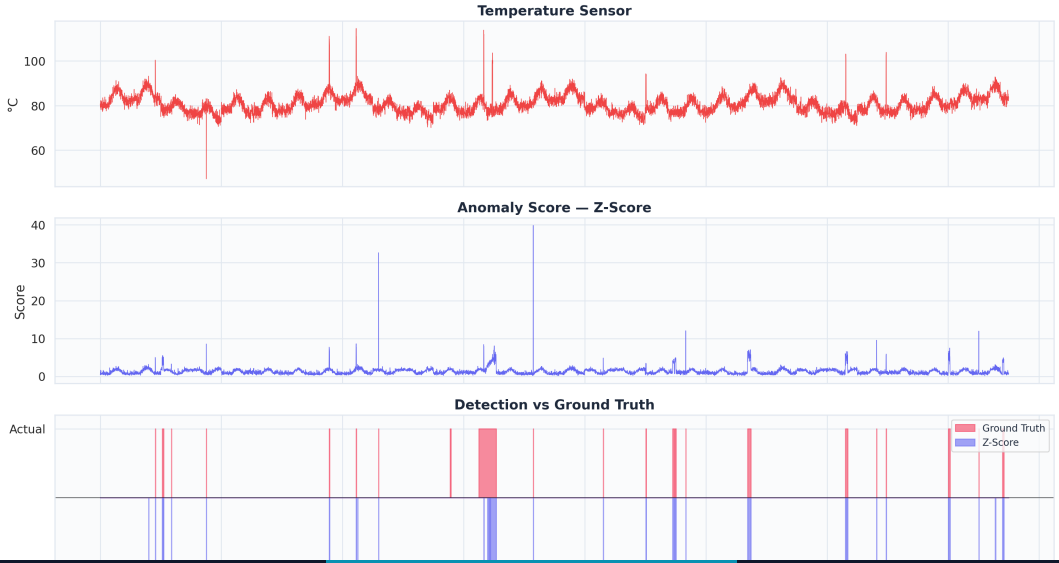
Confusion Matrices



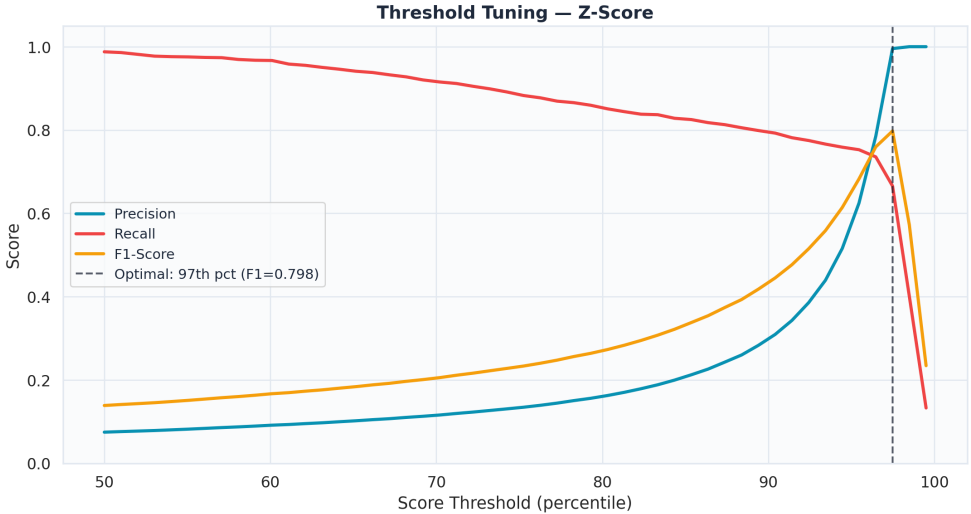
Trade-off

Different methods make different errors. Choosing depends on cost of false positives (alarm fatigue) vs false negatives (missed failures).

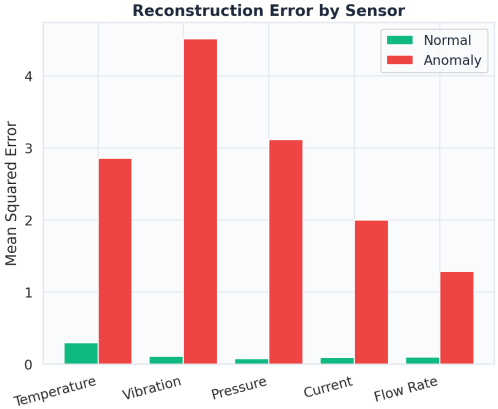
Detection Timeline vs Ground Truth

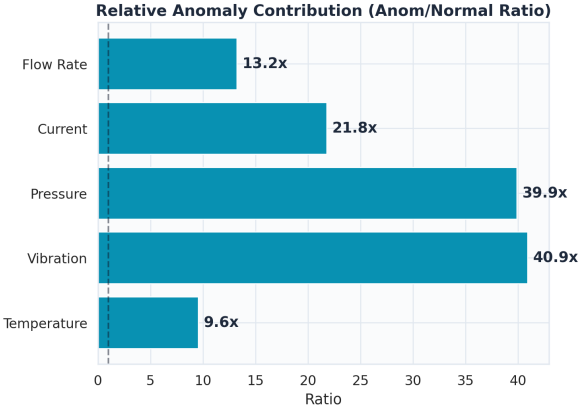


Threshold Tuning



Root-Cause Attribution





Per-Sensor Contribution

PCA reconstruction error per-feature reveals **which sensor caused** the anomaly. This transforms an alert into actionable diagnostic information for the maintenance team.

Executive Summary

IoT Anomaly Detection — Executive Summary

Method Leaderboard

	Precision	Recall	F1	AUC-ROC	AUC-PR
Z-Score	0.994	0.675	0.804	0.944	0.790
IQR	0.780	0.767	0.774	0.744	0.473
EWMA	0.887	0.039	0.074	0.519	0.071
Isolation Forest	0.420	0.611	0.497	0.943	0.533
LOF	0.186	0.249	0.213	0.621	0.242
One-Class SVM	0.463	0.746	0.571	0.939	0.759
PCA Reconstruction	0.542	0.767	0.635	0.930	0.788

```
DETECTION SUMMARY

Monitoring Period:
   30 days (43,200 minutes)

Sensors Monitored:
   5 (Temp, Vib, Pres, Curr, Flow)

Total Anomaly Events:  48
Anomaly Minutes:       1,629
Anomaly Rate:          3.77%

BEST METHOD: Z-Score
   Precision:  0.994
   Recall:     0.675
   F1-Score:   0.804
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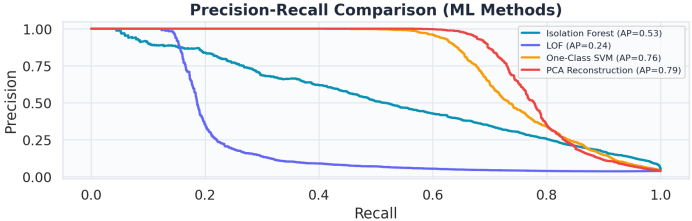

Anomaly Types: Frequency vs Duration

Production Recommendations & Future Work

Production Architecture

1. **Tiered alerting**: Z-Score for high-confidence alerts → PCA for review queue
2. Per-sensor adaptive thresholds
3. Per-feature attribution for diagnostics
4. Time-aware features (hour, day, season)
5. Streaming inference (Kafka + Flink)

Future Work

1. **LSTM Autoencoder** for temporal dynamics
2. **VAE** for probabilistic anomaly scoring
3. Online learning for concept drift
4. Multi-machine fleet transfer learning
5. Predictive maintenance (time-to-failure)
6. Active learning to refine labels

Thank You

Questions & Discussion

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